Nathan Xia

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Engineering Portfolio: Nathan Xia

Education

- **University of Toronto** — Toronto, Canada
  BASc, Engineering Science (Aerospace Engineering Option) — *Sep 2025 – Present*
  cGPA: 3.95/4.00
  Relevant Courses: Vector Calculus and Fluid Mechanics, Thermodynamics and Heat Transfer, Ordinary Differential Equations, Linear Algebra

Projects

- **Lattice Boltzmann CFD Solver** | Ongoing | View Results
  - Built a **D2Q9 lattice Boltzmann CFD solver** from scratch in Python with configurable geometries and flow conditions, velocity/vorticity visualization, and aerodynamic force extraction.
  - Ported core routines to **C++**, achieving approximately **15× faster execution** than the Python implementation.
  - Benchmarked cylinder flow at $Re = 150$: C_d **within 2.2% of a published reference**, with von Kármán vortex shedding and periodic lift and drag fluctuations.
  - Benchmarked NACA 0012 flow at $Re = 2000$, $\alpha = 8°$: **Strouhal number within 1.8% of an ANSYS Fluent reference**.
- **Vortex Generator Research Project** | Paper
  - Designed and conducted a NACA 2412 vortex-generator study in a self-designed and manufactured flow-visualization wind tunnel.
  - Optimized gap width and chordwise placement; regression predicted a **76% increase in the angle at which separation reached 30% chord**.
  - Presented at CYSF and CWSF; earned a **bronze medal among approximately 350 projects**, after selection as 1 of 15 from approximately 600.
- **ANSYS Fluent Analysis – Vortex Generator Project Extension** | View Details
  - Performed **3D ANSYS Fluent simulations** at $Re \approx 4{,}300$, reproducing wind-tunnel geometry and no-slip boundaries.
  - Built locally refined polyhedral meshes with boundary-layer inflation; monitored residuals, force histories, and mass balance.
  - Conducted a **two-mesh sensitivity study**: differences of **0.9% in C_l, 1.4% in C_d, and 1.3 mm in separation location**.
  - Compared baseline and vortex-generator separation using signed wall shear stress; investigated confinement effects on force coefficients.

Experience

- **U of T Institute for Aerospace Studies** — FCET Research Group
  Research Intern — *May 2026 – Sep 2026*
  - Developed a **MATLAB** acquisition and signal-processing pipeline for remote microphone probe calibration, reducing **transfer-function magnitude standard deviation by 69%** and extending the usable frequency range to **1.5×** its original range.
  - Independently managed apparatus and test-matrix design, acquisition, and **FFT-based analysis** across 16 variables; generated a **70+ GB dataset with 1,200+ files**.
  - Validated calibration improvements through **20+ trials**, establishing configurations and procedures for ongoing aeroacoustics research.
  - Selected as **1 of 2 first-year students** to conduct research at UTIAS. Preprint.
- **Robotics for Space Exploration – Aerial** — University of Toronto
  Mechanical Co-Lead — *Jul 2026 – Present*
  - Co-lead mechanical design and integration, coordinating subsystem architecture, CAD, analysis, manufacturing, technical reviews, and mechanical recruits.
  - Planning **CFD parameter studies** of deployable solar panels to build an aerodynamic database for **PID-based descent control and rotational stabilization**.
  - Planning **FEA-based satellite-frame mass optimization**, component CAD, and **CFD verification of parachute performance**.

  Mechanical Sub-team Recruit — *Sep 2025 – Jul 2026*
  - Conceived an airfoil-profile lifting-body glider configuration; evaluated **20+ airfoils in OpenVSP/VSPAERO** to select the geometry.
  - Integrated vortex generators and end plates, achieving **5× the simulated lift at 20 m/s** versus the baseline without these devices.
  - Designed and evaluated **30+ rocket nose-cone profiles** using Autodesk CFD and rocket-performance simulations.

Skills Summary

- **Tools:** ANSYS Fluent, ANSYS Mechanical (FEA), Autodesk CFD, SolidWorks, Fusion 360, Autodesk Inventor, OpenRocket, OpenVSP
- **Skills:** CFD simulation and meshing, aerodynamic analysis and design, wind-tunnel testing, FEA, CAD/CAM, signal processing and data analysis
- **Languages:** Python, C++, MATLAB, C, Java, SystemVerilog